

- KiCad (Schematic Design)
- Tinkercad (Circuit Simulation)

- [GitHub \(Project Repository\)](#)

### Circuit Description:

The traffic light controller circuit consists of a NE555 timer IC, a 74HC4017 decade counter IC, Three LEDs (Red, Yellow, and Green ), resistors, capacitors, and a potentiometer. The NE555 timer is configured in astable mode to generate continuous clock pulses. These pulses are applied to the clock input of the 74HC4017, which sequentially activates its output pins. The outputs are connected to the Red, Yellow and Green LEDs , causing them to light one after another. The potentiometer and timing capacitor determine the pulse frequency, allowing the switching speed of the traffic lights to be adjusted. The entire circuit operates from a regulated 5V DC power supply, provided by the Arduino UNO in this setup.

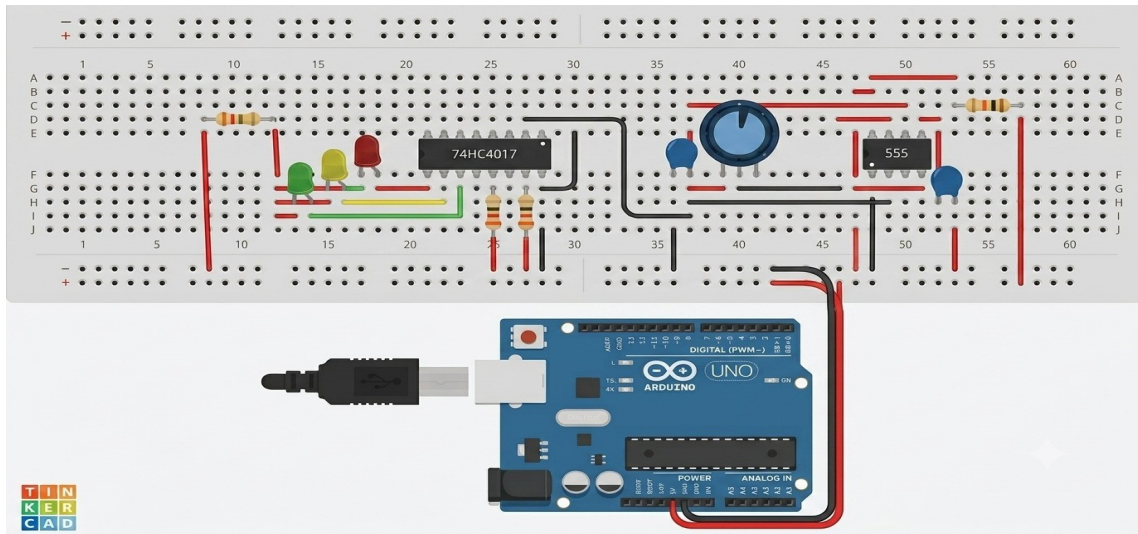


Figure: Tinkercad Simulation of Traffic light controller using timer ic(555)

### Working Principle

The NE555 timer operates in astable mode to generate continuous clock pulses. These pulses are sent to the 74HC4017 decade counter, which sequentially activates the Green, Yellow, and Red LEDs. A potentiometer adjusts the timing by changing the RC time constant, and after completing the sequence, the counter resets and repeats the cycle automatically.

### Result:

The circuit successfully produced an automatic traffic light sequence. The LEDs switched in the correct order, and the delay between each light could be adjusted using the potentiometer while operating on a 5V power supply.

### Conclusion:

This project demonstrates the use of the NE555 timer and 74HC4017 counter for automatic sequential control. It provides practical knowledge of timing circuits and digital electronics and serves as a foundation for developing more advanced traffic control systems.